THE VIRTUAL MEET EXPERIENCE

2017 - 2018

HS VIRTUAL CHALLENGE MEET #4

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2017-2018 Virtual Challenge Meet #4 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.x**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is $77_8 + 113_{16}$?

A. 101010010_2 B. 338_{10} C. 522_8 D. 152_{16} E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. 11 B. -3

C. 5 D. 0

E. There is no output due to an error.

```
out.println(14%(2+3-8*1+2)-6/7);
```

QUESTION 3

What is output by the code to the right?

A. you\tare\n\t\"\'fake\tnews

B. you are "'fake news

C. you are
"'fake news

D. you are

"'fake news

E. There is no output due to an error.

```
out.print("you\t");
out.print("are\n\t");
out.println("\"\'fake\tnews");
```

QUESTION 4

What is output by the code to the right?

A. itsgonnabehugeitzgonnabehuge

B. itzgonnabehuge

C. itsgonnabehuge

D. itsgonnabehuge itzgonnabehuge

E. There is no output due to a syntax error.

```
String s="itsgonnabehuge";
String k=s.replace("s", "z");
s.concat(k);
out.print(s);
```

QUESTION 5

What is output by the code to the right?

A. true B. false

```
boolean a=false;
boolean b=!a||false^true;
a=a&&b&&(a^b^!a);
out.println(a&&b||a);
```

QUESTION 6

What is output by the code to the right?

A. 4.0 5.0 5 B. 4 5 4

C. 4.0 5.0 4 D. 4 5 5

E. There is no output due to an error.

```
double k=4.5;
out.print(Math.floor(k)+" ");
out.print(Math.ceil(k)+" ");
out.print(Math.round(k)+" ");
```

QUESTION 7

What is the output by the code to the right?

A. `11`
B. `10`
C. `10.6`
D. `5.6`
E. There is no output due to an error.

```
short k=5;
k+=5.6;
out.print(k);
```

QUESTION 8

What is the output by the code to the right?

A. `2123`
B. `212`
C. `1212`
D. `13`
E. There is no output due to an error.

```
String a="China",
     b=a.toUpperCase().toLowerCase();
if(a.equals(b))
 out.print(1);
if(a.contains(b.substring(1)))
 out.print(2);
 out.print(12);
else if(a.charAt(1)==b.charAt(2))
 out.print(3);
```

QUESTION 9

What is the output by the code to the right ?

A. `363`
B. `120`
C. `1092`
D. `0`
E. There is no output due to a syntax error.

```
int k=5,tot=0,num=3;
do {
 k--;
 tot+=num;
 num*=3;
}while(k!=0);
out.print(tot);
```

QUESTION 10

What is the output by the code to the right ?

A. `[2, 4, 1, 3, 1, 1, 2, 5]`
B. `[2, 6, 1, 3, 1, 0, 2, 5]`
C. `[2, 6, 4, 3, 1, 0, 2, 5]`
D. There is no output due to a runtime error
E. Output cannot be determined until runtime.

```
int[] k= {2,6,4,3,1,0,2,5};
k[1]=k[k[k[2]]];
k[k[k[2]]]=k[k[4]];
k[1+k[4]]=k[4];
out.print(Arrays.toString(k));
```

QUESTION 11

Which of the following could read in the following data?

```
6.0
```

A. `file.nextInt()`
B. `file.hasNextDouble()`
C. `file.nextLine()`
D. More than one of the above.
E. None of the above.

QUESTION 12

What is the output by the code to the right ?

A. `430`
B. `862`
C. `660`
D. `860`
E. There is no output due to a syntax error.

```
int t=2;
for(int y=0;y<10;y++)
 for(int x=y;x<11;x++)
   for(int h=0;h<2;h++)
     t+=x;
out.print(t);
```

QUESTION 13

Which of the operations to the right has the highest precedence?

A. III

B. IV

C. I

D. II

E. V

```
I. |(logical)
II. ==
III. ++
IV. instanceof
V. type cast
```

QUESTION 14

What is the output by the code to the right ?

A. 4

B. 64

C. 8

D. 32

E. 16

```
out.print(Float.SIZE);
```

QUESTION 15

What is the output by the code to the right ?

A. [2, 1, 0]

B. [4, 2, 1]

C. []

D. [4, 2, 0]

E. There is no output due to an error.

```
ArrayList<Integer>a;
a=new ArrayList<Integer>();
a.add(4);
a.add(2);
a.add(1);
a.add(0);
a.remove(4);
out.print(a);
```

QUESTION 16

What is the output by the code to the right ?

A. 261

B. 237

C. 262

D. 238

E. There is no output due to a syntax error.

```
out.print(32+4>>2|212+17);
```

QUESTION 17

What is the output by the code to the right ?

A. 23

B. 22

C. 41

D. 40

E. There is no output due to an error.

```
int t=1;
for(int y=0;y<=10;y+=3) {
 for(int x=0;x<=y*2;x++)
    t+=1;
}
System.out.println(t);
```

QUESTION 18

What is output by the line marked `//q18` code to the right?

A. `true`

B. `false`

C. `212`

D. There is no output due to a runtime error.

E. There is no output due to a syntax error.

QUESTION 19

What is output by the line marked `//q19` code to the right?

A. `[212, f]`

B. `[1, 17, AS]`

C. `[2, 34, USA]`

D. There is no output due to a runtime error.

E. There is no output due to a syntax error.

```
TreeSet<String>a,b;
a=new TreeSet<String>();
b=new TreeSet<String>();
a.add(""+1);a.add(""+212);
a.add(""+17);a.add("AS");
a.add("f");
b.add("34");b.add("USA");
b.add("212");b.add("f");
b.add("2");
out.println(a.retainAll(b));  //q18
out.print(a);                 //q19
```

QUESTION 20

What is the big O runtime of adding a value to a binary search tree?

A. O(n)

B.O(n^2)

C.O(log(n))

D.O(n log(n))

E. O(n*n/3)

QUESTION 21

Which sorting algorithm was implemented in `Arrays.sort()`?

A. Selection sort

B. Insertion sort

C. Quick sort

D. Merge sort

E. Spaghetti sort

```
String y="a 3 8 HN 3 JK y hg 8 *& y";
String[]s=y.split(" ");
Arrays.sort(s);
```

```
// questions 22-25 refer to class Athlete

class Athlete <1*>{
  int speed, jump, strong, agile;
  String name;
  public Athlete(String n)
     {name=n;speed=jump=strong=agile=12;}
  public int compareTo(Athlete a) {
     int t=meet();
     int ta=a.meet();
     return t==ta?name.compareTo(a.name):ta-t;
  }
  public int meet()
  {return run(100)+shot(10)+hjump(6)+hurdle(100);}
  public int run(int dist) {return dist/speed;}
  public int shot(int lbs) {return 2*strong/lbs;}
  public int hjump(int h) {return jump>h?10:2;}
  public int hurdle(int dist) {return dist/agile;}
  public String toString() {return name;}
}
class Lebron <2*>{
  int dunk;
  public Lebron(String n) {super(n);dunk=12;}
  public int compareTo(Athlete a) {
     if(a instanceof Lebron && ((Lebron)a).dunk!=dunk)
        return dunk;
     return super.compareTo(a);
  }
  public int dunker() {return dunk*5;}
  public String toString() {return name+" "+dunker();}
}

////////////client code//////////////////////////////////////////////////////////////
ArrayList<Athlete>a=new ArrayList<Athlete>();
a.add(new Athlete("CP3"));
a.get(0).agile-=3;
a.add(new Athlete("Springer"));
a.get(1).strong+=3;
a.add(new Athlete("Altuve"));
a.get(2).speed-=3;
a.add(new Athlete("Harden"));
a.get(3).agile-=2;
a.get(3).jump+=2;
a.add(new Athlete("Watt"));
a.get(4).strong+=4;
a.add(new Lebron("Lebron"));
a.get(5).jump=14;
((Lebron)a.get(5)).dunk+=3;
a.add(new Lebron("Jebron"));
a.get(6).speed+=212;
Collections.sort(a);
out.println(a.get(1)+" "+a);  //1
```

QUESTION 22

```
// questions 22-25 refer to class Athlete from the prior page
```

Which of the following could replace **<1*>** in the code above, so the client code will compile and run correctly?

A. `implements Comparable`

B. `extends Comparable`

C. `implements Comparable<Athlete>`

D. More than one of the above will work.

E. No additional code is needed.

QUESTION 23

Assuming that **<1*>** is filled correctly, which of these could replace **<2*>** in the code above so that class `Lebron` will work as intended?

A. `implements Comparable<Lebron>`

B. `extends Athlete`

C. `extends Comparable, Athlete`

D. `implements Comparable, Athlete`

E. `implements Comparable`

QUESTION 24

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by the the client code to the right?

A. `Kobe`

B. `KobeShaq`

C. `Shaq`

D. Output cannot be determined until runtime.

E. This is no output due to an error.

```
Athlete a=new Athlete("Kobe");
Athlete b=new Athlete("Shaq");
a.speed-=3;
a.strong-=2;
a.agile-=3;
b.speed+=2;
b.jump+=2;
b.strong+=5;
b.agile+=2;
out.println(a.compareTo(b)>0?a:b);
```

QUESTION 25

Assuming that **<1*>** is filled correctly, what is the output by the client code marked `//1`?

A. `Lebron [Jebron, Lebron, Watt, Springer, Harden, CP3, Altuve]`
B. `Lebron 75 [Jebron 60, Lebron 75, Watt, Springer, Harden, CP3, Altuve]`
C. `CP3 [Altuve, CP3, Harden, Springer, Watt, Lebron, Jebron]`
D. `CP3 [Altuve, CP3, Harden, Springer, Watt, Lebron 75, Jebron 60]`
E. This is no output due to an error.

QUESTION 26

What is output by the code to the right?

A. `[4, 5, 6, 7, 8]`

B. `[7, 7, 7, 7]`

C. Output cannot be determined until runtime.

D. There is no output due to an error.

E. None of the above.

```
int[] k= {4,5,6,7,8};
k= new int[]{7,7,7,7};
out.print(k);
```

QUESTION 27

Which of the following is not a valid java identifier?

A. `o00o`

B. `for`

C. `_$$_`

D. `String`

E. More than one of the above is invalid.

QUESTION 28

What is output by the line marked `//q28`?

A. `71`

B. `72`

C. `73`

D. `74`

E. `75`

QUESTION 29

What is output by the line marked `//q29`?

A. `71`

B. `72`

C. `73`

D. `74`

E. `75`

QUESTION 30

What is output by the line marked `//q30`?

A. `91`

B. `92`

C. `93`

D. `94`

E. `0`

```
public static int recur(int y)
{
 if(y<20)
    return recur(y+2)+1;
 if(y<85)
    return recur(y+2)+2;
 else
    return 1;
}


//////client code//////////////////
out.println(recur(4));  //q28
out.println(recur(1));  //q29
out.print(recur(-31));  //q30
```

```
//questions 31-34 refer to the following structure
class Structure<E>{
  Node root;
  public Structure(<1*> o) {
          root=new Node(o);
  }
  public void add(<1*> o) {
          Node n=root;
          while(n!=null) {
            if(n.data.compareTo(o)>0) {
                  if(n.rt==null) {n.rt=new Node(o);n=null;}
                  else n=n.rt;
            }
            else {
                  if(n.lt==null) {n.lt=new Node(o);n=null;}
                  else n=n.lt;
            }
          }
  }
  public Node get() {
          Node n=root;
          while(n.lt!=null)n=n.lt;
          return n;
  }
}
class Node{
  Node lt,rt;
  <1*> data;
  public Node(<1*> c) {
          data=c;
  }
  public String toString() {
          return ""+data;
  }
}

/////client code/////////////////////
Structure<String>a = a=new Structure<String>("AAA");
a.add("YYY");
a.add("212");
a.add("111");
a.add("dfg");
a.add("ioj");
a.add("abc");
a.add("ABC");
out.println(a.get()); //1
Node n1=new Node(1);
Node n2=new Node(2);
Node n3=new Node(3);
n1.lt=n2;
n2.lt=n3;
n3.lt=n1;
n1.rt=n1.lt.lt;
n2.rt=n1.rt.lt;
n3.rt=n3.lt;
Node n=n1;
for(int y=1;y<=5000;y*=2)
  n=y<200?n.rt:n.lt;
out.println(n.data);  //2
```

QUESTION 31

```
//questions 31-34 refer to the structure on the prior page
```

What could replace **<1*>** in the code to the right so that the `Structure` class works as intended?

A. `Object`
B. `String`
C. `Comparable`
D. All of the above.
E. B and C

QUESTION 32

What is the output by the line in the client code marked `//1`?

A. `111`
B. `AAA`
C. `ioj`
D. There is no output due to an error.
E. `ABC`

QUESTION 33

What is the output by the line in the client code marked `//2`?

A. `3`
B. `2`
C. `1`
D. There is no output due to an error.
E. Output cannot be determined until runtime.

QUESTION 34

What data structure is implemented by the `Structure` class?

A. Binary Search Tree
B. min heap
C. max heap
D. Priority Queue
E. Hashtable

QUESTION 35

What is output by the code to the right?

A. `0`
B. `2147483648`
C. `2147483647`
D. `-2147483648`
E. This is no output due to a syntax error.

```
out.print(Integer.MAX_VALUE+1);
```

QUESTION 36

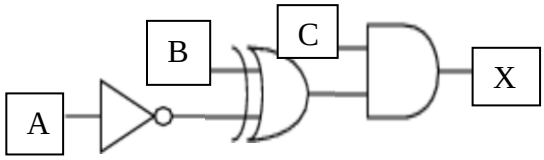
What boolean expression is represented to the right ?

A. (!A || B) && C
B. (!A ^ B) && C
C. !A ^ B || C
D. !(A ^ B && C)
E. A ^ B && C

QUESTION 37

How many ordered triples will cause boolean X in the expression in the right to be assigned the value false ?

A. `4`
B. `5`
C. `3`
D. `2`
E. `6`



QUESTION 38

What is the big O runtime of an average-case quicksort?

QUESTION 39

What is the value of the following postfix expression?

```
3 5 + 7 - 8 * 17  3 / 2 3 ^ - 2 * +
```

QUESTION 40

What would a min heap look like if you inserted the following values in order (level-order)?

```
3 12 212 2 84 1 99 17 2
```

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2017-2018 Virtual Challenge Meet #4
Computer Science - KEY

1) E
2) D
3) C
4) C
5) B
6) A
7) B
8) E
9) A
10) B

11) C
12) B
13) A
14) D
15) E
16) B
17) C
18) A
19) A
20) C

21) D
22) C
23) B
24) C
25) D
26) C
27) B
28) E
29) E
30) A

31) C
32) C
33) A
34) D
35) D
36) B
37) E
38) O(N*logN)
39) 2
40) [1, 2, 2, 3, 84, 212, 99, 17, 12]

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors.**
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- Assume any undefined (undeclared) variables have been defined as used.

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